INFOSYS SE PSEUDOCODE VAULT • LOOPS & CONDITIONAL TRACING

While, For, Nested Loops, Break/Continue, Step Modifications • Total Solved Questions: 74

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: For Loop with Condition

Q1. What will be the output of the following pseudocode?

```
Set Integer s = 0
for i = 1 to 10 step 1 do
if (i % 3 != 0)
s = s + i
end-if
end-for
display s
```

(A) 30
(B) 37
(C) 42
(D) 45

Correct Answer: Option (B) : 37

■ **Step-by-Step Logic:** Total 1..10=55; multiples of 3 are 3+6+9=18; 55–18=37.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Nested Loop

Q2. What will be the output of the following pseudocode?

```
Set Integer c = 0
for i = 1 to 4 step 1 do
for j = 1 to 5 step 1 do
if (i + j == 6)
c = c + 1
end-if
end-for
end-for
display c
```

(A) 2
(B) 3
(C) 4
(D) 5

Correct Answer: Option (C) : 4

■ **Step-by-Step Logic:** Pairs are (1,5),(2,4),(3,3),(4,2): 4.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Nested Loop Sum

Q3. What will be the output of the following pseudocode?

```
Set Integer s = 0
for i = 1 to 3 step 1 do
for j = 1 to i + 1 step 1 do
s = s + j
end-for
end-for
display s
```

(A) 17
(B) 18
(C) 19
(D) 20

Correct Answer: Option (C) : 19

■ **Step-by-Step Logic:** For i=1,2,3 the inner sums are 3,6,10; total=19. Infosys Medium Pseudocode Practice • Page 36

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: While Loop

Q4. What will be the output of the following pseudocode?

```
Set Integer x = 5
Set Integer s = 0
while (x <= 25)
s = s + x
x = x + 5
end-while
display s
```

(A) 50
(B) 60
(C) 75
(D) 80

Correct Answer: Option (C) : 75

■ **Step-by-Step Logic:** Execution Trace: 1. Initialize loop variables and verify loop condition on each pass. 2. Execute loop body and apply state modifications. 3. Terminate when the guard condition evaluates to false. 4. The final variable state upon loop termination produces Option (C) (75).

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Break in Nested Loop

Q5. What will be the output of the following pseudocode?

```
Set Integer c = 0
for i = 1 to 4 step 1 do
for j = 1 to 4 step 1 do
if (j == 3)
break
end-if
c = c + 1
end-for
end-for
display c
```

(A) 4
(B) 6
(C) 8
(D) 12

Correct Answer: Option (C) : 8

■ **Step-by-Step Logic:** The inner loop adds 1 for j=1 and j=2, then breaks at j=3. 4×2=8.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Continue

Q6. What will be the output of the following pseudocode?

```
Set Integer s = 0
for i = 1 to 8 step 1 do
if (i % 3 == 0)
continue
end-if
s = s + i
end-for
display s
```

(A) 24
(B) 27
(C) 30
(D) 33

Correct Answer: Option (B) : 27

■ **Step-by-Step Logic:** 1..8 sum=36; skip 3 and 6, so 36–9=27.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Reverse Loop

Q7. What will be the output of the following pseudocode?

```
Set Integer s = 0
for i = 12 to 2 step -2 do
s = s + i / 2
end-for
display s
```

(A) 18
(B) 21
(C) 24
(D) 27

Correct Answer: Option (B) : 21

■ **Step-by-Step Logic:** Values of i/2: 6,5,4,3,2,1; sum=21.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Loop with Changing Step

Q8. What will be the output of the following pseudocode?

```
Set Integer i = 1
Set Integer s = 0
while (i < 20)
s = s + i
i = i * 2 + 1
end-while
display s
```

(A) 24
(B) 26
(C) 28
(D) 30

Correct Answer: Option (B) : 26

■ **Step-by-Step Logic:** i values are 1,3,7,15; sum=26.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Nested Loop Count

Q9. What will be the output of the following pseudocode?

```
Set Integer c = 0
for i = 1 to 5 step 1 do
for j = i to 5 step 1 do
c = c + 1
end-for
end-for
display c
```

(A) 10
(B) 12
(C) 15
(D) 20

Correct Answer: Option (C) : 15

■ **Step-by-Step Logic:** Counts are 5+4+3+2+1=15.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Loop with Modulo

Q10. What will be the output of the following pseudocode?

```
Set Integer s = 0
for i = 2 to 14 step 2 do
if (i % 4 == 0)
s = s + i
end-if
end-for
display s
```

(A) 12
(B) 16
(C) 20
(D) 24

Correct Answer: Option (D) : 24

■ **Step-by-Step Logic:** Multiples of 4 are 4,8,12; sum=24.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Nested Conditional Loop

Q11. What will be the output of the following pseudocode?

```
Set Integer s = 0
for i = 1 to 5 step 1 do
if (i % 2 == 1)
for j = 1 to i step 2 do
s = s + j
end-for
end-if
end-for
display s
```

(A) 12
(B) 14
(C) 16
(D) 18

Correct Answer: Option (B) : 14

■ **Step-by-Step Logic:** For i=1,3,5 the inner sums are 1,4,9; total=14. Infosys Medium Pseudocode Practice • Page 37

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Loop State

Q12. What will be the output of the following pseudocode?

```
Set Integer x = 3
for i = 1 to 4 step 1 do
x = x + i
x = x - 1
end-for
display x
```

(A) 8
(B) 9
(C) 10
(D) 11

Correct Answer: Option (B) : 9

■ **Step-by-Step Logic:** Each iteration changes x by i–1. Total change=0+1+2+3=6; final x=9.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Loop Product

Q13. What will be the output of the following pseudocode?

```
Set Integer p = 1
for i = 2 to 8 step 2 do
p = p * (i / 2)
end-for
display p
```

(A) 12
(B) 24
(C) 36
(D) 48

Correct Answer: Option (B) : 24

■ **Step-by-Step Logic:** Factors are 1,2,3,4; product=24.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: While + Modulo

Q14. What will be the output of the following pseudocode?

```
Set Integer n = 29
Set Integer c = 0
while (n > 0)
if (n % 2 == 1)
c = c + 1
end-if
n = n / 2
end-while
display c
```

(A) 2
(B) 3
(C) 4
(D) 5

Correct Answer: Option (B) : 3

■ **Step-by-Step Logic:** Binary 29 is 11101, containing four 1s.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Nested Loop Arithmetic

Q15. What will be the output of the following pseudocode?

```
Set Integer s = 0
for i = 1 to 3 step 1 do
for j = 1 to 3 step 1 do
s = s + i * j
end-for
end-for
display s
```

(A) 12
(B) 18
(C) 24
(D) 36

Correct Answer: Option (D) : 36

■ **Step-by-Step Logic:** The double sum factorizes: (1+2+3)×(1+2+3)=6×6=36.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Break with Sum

Q16. What will be the output of the following pseudocode?

```
Set Integer s = 0
for i = 2 to 12 step 2 do
s = s + i
if (s > 15)
break
end-if
end-for
display s
```

(A) 12
(B) 18
(C) 20
(D) 24

Correct Answer: Option (B) : 18

■ **Step-by-Step Logic:** Running sums: 2,6,12,20; break after 20.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Continue with Product

Q17. What will be the output of the following pseudocode?

```
Set Integer p = 1
for i = 1 to 6 step 1 do
if (i % 2 == 0)
continue
end-if
p = p * i
end-for
display p
```

(A) 9
(B) 12
(C) 15
(D) 20

Correct Answer: Option (C) : 15

■ **Step-by-Step Logic:** Odd values 1,3,5 are multiplied: 1×3×5=15.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Loop with Shift

Q18. What will be the output of the following pseudocode?

```
Set Integer s = 0
for i = 1 to 4 step 1 do
s = s + (i << 1)
end-for
display s
```

(A) 16
(B) 18
(C) 20
(D) 24

Correct Answer: Option (C) : 20

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (C) (20).

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Loop with Right Shift

Q19. What will be the output of the following pseudocode?

```
Set Integer s = 0
for i = 8 to 32 step 8 do
s = s + (i >> 2)
end-for
display s
```

(A) 15
(B) 18
(C) 20
(D) 24

Correct Answer: Option (C) : 20

■ **Step-by-Step Logic:** The shifted values are 2,4,6,8; sum=20. Infosys Medium Pseudocode Practice • Page 38

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Nested Loop with Break

Q20. What will be the output of the following pseudocode?

```
Set Integer c = 0
for i = 1 to 4 step 1 do
for j = 1 to 4 step 1 do
c = c + 1
if (c == 7)
break
end-if
end-for
end-for
display c
```

(A) 7
(B) 8
(C) 12
(D) 13

Correct Answer: Option (D) : 13

■ **Step-by-Step Logic:** The counter continues across outer iterations. It reaches 7, breaks the current inner loop, then continues; 4 more inner iterations occur, so final c=13.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Loop and Comparison

Q21. What will be the output of the following pseudocode?

```
Set Integer c = 0
for i = 1 to 9 step 1 do
if (i * i < 50)
c = c + 1
end-if
end-for
display c
```

(A) 6
(B) 7
(C) 8
(D) 9

Correct Answer: Option (B) : 7

■ **Step-by-Step Logic:** i=1..7 satisfy $i^2<50$; count=7.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Function with Loop

Q22. What will be the output of the following pseudocode?

```
Integer f(Integer n)
Set Integer s = 1
for i = 1 to n step 1 do
s = s + i
end-for
return s
End function
display f(6)
```

(A) 20
(B) 21
(C) 22
(D) 23

Correct Answer: Option (C) : 22

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields Option (C) (22).

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Function Condition

Q23. What will be the output of the following pseudocode?

```
Integer f(Integer n)
if (n % 4 == 0)
return n / 2
else
return n + 4
end-if
End function
display f(14)
```

(A) 9
(B) 14
(C) 18
(D) 28

Correct Answer: Option (C) : 18

■ **Step-by-Step Logic:** 14 is not divisible by 4, so 14+4=18. Infosys Medium Pseudocode Practice • Page 42

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Function Digit Logic

Q24. What will be the output of the following pseudocode?

```
Integer f(Integer n)
Set Integer s = 0
while (n > 0)
s = s + n % 10
n = n / 10
end-while
return s
End function
display f(386)
```

(A) 14
(B) 15
(C) 16
(D) 17

Correct Answer: Option (D) : 17

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields Option (D) (17).

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Function Nested

Q25. What will be the output of the following pseudocode?

```
Integer g(Integer x)
return x * 2
End function
Integer f(Integer x)
return g(x) + g(x+1)
End function
display f(4)
```

(A) 16
(B) 18
(C) 20
(D) 22

Correct Answer: Option (B) : 18

■ **Step-by-Step Logic:** g(4)=8 and g(5)=10; total=18.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Function Loop Product

Q26. What will be the output of the following pseudocode?

```
Integer f(Integer n)
Set Integer p = 1
for i = 1 to n step 1 do
if (i % 2 == 0)
p = p * i
end-if
end-for
return p
End function
display f(6)
```

(A) 24
(B) 36
(C) 48
(D) 60

Correct Answer: Option (C) : 48

■ **Step-by-Step Logic:** Multiply even values 2×4×6=48.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Function with Loop and Shift

Q27. What will be the output of the following pseudocode?

```
Integer f(Integer n)
Set Integer s = 0
for i = 1 to n step 1 do
s = s + (i << 1)
end-for
return s
End function
display f(5)
```

(A) 20
(B) 25
(C) 30
(D) 35

Correct Answer: Option (C) : 30

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (C) (30).

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Mixed Loop + XOR

Q28. What will be the output of the following pseudocode?

```
Set Integer x = 0
for i = 1 to 5 step 1 do
x = x ^ i
end-for
display x
```

(A) 0
(B) 1
(C) 2
(D) 3

Correct Answer: Option (A) : 0

■ **Step-by-Step Logic:** 0^1=1, ^2=3, ^3=0, ^4=4, ^5=1.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Mixed Loop + AND

Q29. What will be the output of the following pseudocode?

```
Set Integer x = 15
for i = 1 to 3 step 1 do
x = x & (14 - i)
end-for
display x
```

(A) 8
(B) 10
(C) 12
(D) 14

Correct Answer: Option (B) : 10

■ **Step-by-Step Logic:** x=15&13=13; 13&12=12; 12&11=8.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Mixed Nested Loop + Modulo

Q30. What will be the output of the following pseudocode?

```
Set Integer c = 0
for i = 1 to 4 step 1 do
for j = 1 to 4 step 1 do
if ((i + j) % 3 == 0)
c = c + 1
end-if
end-for
end-for
display c
```

(A) 4
(B) 5
(C) 6
(D) 7

Correct Answer: Option (B) : 5

■ **Step-by-Step Logic:** Pairs with sum divisible by 3: 5 pairs. Correct answer B.

[OnlineStudy4U 100 Medium-Level MCQs]

Topic: Mixed Nested Conditions

Q31. What will be the output of the following pseudocode?

```
Set Integer a = 9
Set Integer b = 6
Set Integer c = 3
if (a > b)
if ((a - b) == c)
display a + b + c
else
display a - b
end-if
else
display b + c
end-if
```

(A) 12
(B) 15
(C) 18
(D) 21

Correct Answer: Option (C) : 18

■ **Step-by-Step Logic:** 9>6 and 9–6=3, so the nested condition is true; 9+6+3=18.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q32. How many times does the loop body run, and

```
what is printed?
Integer
i, count
Set
i = 1, count = 0
while
i < 20 do
i
= i * 2
count
= count + 1
end
while
Print
count
```

(A) 3
(B) 4
(C) 5
(D) 6

Correct Answer: Option (C) : 5

■ **Step-by-Step Logic:** Trace i and count: start i=1. Pass 1: i=2, count=1. Pass 2: i=4, count=2. Pass 3: i=8, count=3. Pass 4: i=16, count=4. Pass 5: i=32, count=5. Now i=32 is not less than 20, so the loop stops. Printed value is 5.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q33. What will be the output of the following pseudocode?

```
What is the output of the following Pseudo
Code?
Integer
x, i
Set
x = 0
for
i = 1 to 6 do
if
i % 2 == 0 then
continue
end
if
x
= x + i
end
for
Print
x
```

(A) 9
(B) 12
(C) 16
(D) 21

Correct Answer: Option (A) : 9

■ **Step-by-Step Logic:** continue skips even values of i. The loop runs i = 1 to 6, so only the odd values 1, 3 and 5 are added. x = 1 + 3 + 5 = 9.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q34. What will be the output of the following pseudocode?

```
What is printed by the following Pseudo
Code?
Integer
i, j, count
Set
count = 0
for
i = 1 to 4 do
for
j = 1 to i do
count
= count + 1
end
for
end
for
Print
count
```

(A) 4
(B) 8
(C) 10
(D) 16

Correct Answer: Option (C) : 10

■ **Step-by-Step Logic:** The inner loop runs i times for each i from 1 to 4. Total iterations = 1+2+3+4 = 10. So count prints 10.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q35. What will be the output of the following pseudocode?

```
A queue follows FIFO. What is printed by the
following Pseudo Code?
Queue
Q
enqueue(Q,
2)
enqueue(Q,
4)
enqueue(Q,
6)
dequeue(Q)
enqueue(Q,
8)
Print
front(Q)
```

(A) 2
(B) 4
(C) 6
(D) 8

Correct Answer: Option (B) : 4

■ **Step-by-Step Logic:** After three enqueues the queue is [2, 4, 6] with 2 at the front. dequeue removes 2. enqueue 8 makes [4, 6, 8]. front returns 4.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q36. What will be the output of the following pseudocode?

```
A stack follows LIFO. What is printed by the
following Pseudo Code?
Stack
S
push(S,
5)
push(S,
8)
push(S,
3)
pop(S)
push(S,
7)
Print
peek(S)
```

(A) 3
(B) 5
(C) 7
(D) 8

Correct Answer: Option (C) : 7

■ **Step-by-Step Logic:** Stack after the three pushes: [5, 8, 3] with 3 on top. pop removes 3. push 7 makes the stack [5, 8, 7]. peek returns the top element, which is 7.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q37. What will be the output of the following pseudocode?

```
What is printed?
Integer
n, d
Set
n = 48, d = 0
while
n > 1 do
n
= n / 2
d
= d + 1
end
while
Print
d
```

(A) 3
(B) 4
(C) 5
(D) 6

Correct Answer: Option (C) : 5

■ **Step-by-Step Logic:** 48→24→12→6→3→1. That is five divisions. d = 5. (Integer division: 3/2=1.)

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q38. What will be the output of the following pseudocode?

```
What is printed?
Integer
A[5] = {2, 4, 6, 8, 10}
Integer
i, s
Set
s = 0
for
i = 0 to 4 do
if
A[i] % 4 == 0 then s = s + A[i]
end
for
Print
s
```

(A) 12
(B) 18
(C) 24
(D) 30

Correct Answer: Option (A) : 12

■ **Step-by-Step Logic:** Values divisible by 4 are 4 and 8. s = 4 + 8 = 12.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q39. What will be the output of the following pseudocode?

```
What does G(4) return?
Function
G(n)
if
n <= 1 then return 1
return
n * G(n - 1)
End
Function
Print
G(4)
```

(A) 10
(B) 16
(C) 24
(D) 64

Correct Answer: Option (C) : 24

■ **Step-by-Step Logic:** G is factorial. G(4)=4×3×2×1=24.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q40. What will be the output of the following pseudocode?

```
[Loops]
What is printed?
Integer
i, s
Set
s = 0
for
i = 1 to 10 do
if
i % 2 == 1 then s = s + i
end
for
Print
s
```

(A) 25
(B) 30
(C) 45
(D) 55

Correct Answer: Option (A) : 25

■ **Step-by-Step Logic:** The loop adds the odd values 1+3+5+7+9 = 25.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q41. What will be the output of the following pseudocode?

```
[Loops]
What is printed?
Integer
i, p
Set
p = 1
for
i = 1 to 4 do
p
= p * i
end
for
Print
p
```

(A) 10
(B) 16
(C) 24
(D) 120

Correct Answer: Option (C) : 24

■ **Step-by-Step Logic:** p multiplies 1×2×3×4 = 24.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q42. What will be the output of the following pseudocode?

```
What
does R(6) return?
Function
R(n)
if
n < 3 then return n
return
R(n - 1) + 2
End
Function
Print
R(6)
```

(A) 6
(B) 8
(C) 10
(D) 12

Correct Answer: Option (C) : 10

■ **Step-by-Step Logic:** R(2)=2. Each later call adds 2. R(6)=2 + 2×4 = 10.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q43. What will be the output of the following pseudocode?

```
What
is printed?
Integer
i, c
Set
c = 0
for
i = 1 to 8 do
if
i % 2 == 0 then c = c + i
end
for
Print
c
```

(A) 16
(B) 18
(C) 20
(D) 36

Correct Answer: Option (C) : 20

■ **Step-by-Step Logic:** Even values 2+4+6+8 = 20.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q44. What will be the output of the following pseudocode?

```
What
does U(4) return?
Function
U(n)
if
n == 0 then return 1
return
n + U(n - 1)
End
Function
Print
U(4)
```

(A) 8
(B) 9
(C) 11
(D) 24

Correct Answer: Option (C) : 11

■ **Step-by-Step Logic:** U(0)=1, then +1+2+3+4 = 11.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q45. What will be the output of the following pseudocode?

```
What
is printed?
Integer
i, n
Set
n = 0
for
i = 1 to 6 do
if
i % 3 == 0 then n = n + i
end
for
Print
n
```

(A) 6
(B) 9
(C) 12
(D) 21

Correct Answer: Option (B) : 9

■ **Step-by-Step Logic:** Multiples of 3 from 1 to 6 are 3 and 6. Sum = 9.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q46. What will be the output of the following pseudocode?

```
What
is printed?
Integer
i, t
Set
t = 1
for
i = 1 to 5 do
t
= t * 2
end
for
Print
t
```

(A) 10
(B) 16
(C) 32
(D) 64

Correct Answer: Option (C) : 32

■ **Step-by-Step Logic:** Start 1, multiply by 2 five times: 32.

[OnlineStudy4U Master E-Book 4]

Topic: Pseudocode Execution Tracing

Q47. What will be the output of the following pseudocode?

```
What
does W(5) return?
Function
W(n)
if
n <= 1 then return 2
return
W(n - 1) + 3
End
Function
Print
W(5)
```

(A) 11
(B) 14
(C) 15
(D) 17

Correct Answer: Option (B) : 14

■ **Step-by-Step Logic:** W(1)=2. Add 3 four times: 2+12=14.

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q48. 1. What will be the output of the following pseudo code?

```
Input m = 9,n = 6,
m= m + 1;
n= n - 1;
m= m + n;
if(m>n)
print m
else
print n
```

(A) 5
(B) 7
(C) 9
(D) 12

Correct Answer: Option (A) : 5

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches Option (A) (5).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q49. 1. What will be the output of the following pseudocode?

```
Integer i
Set i = 3
do
print i + 3
i = i - 1
while(i not equals 0)
end while
```

(A) 6 6 6
(B) 6 5 6
(C) 5 5 5
(D) 6 5 4

Correct Answer: Option (D) : 6 5 4

■ **Step-by-Step Logic:** Execution Trace: 1. Initialize loop variables and verify loop condition on each pass. 2. Execute loop body and apply state modifications. 3. Terminate when the guard condition evaluates to false. 4. The final variable state upon loop termination produces Option (D) (6 5 4).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q50. What will be the output of the following pseudocode?

```
1.     If the sequence of operations are performed on a
stack, the sequence of pop will be?

Push(1), Push(2), Pop(), Push(1), Push(2), Pop(), Pop(),
Pop(), Push(2), Pop().
```

(A) 2 2 1 1 2
(B) 2 2 1 2 2
(C) 2 1 2 2 1
(D) 2 1 2 2 2

Correct Answer: Option (A) : 2 2 1 1 2

■ **Step-by-Step Logic:** Execution Trace: 1. Initialize loop variables and verify loop condition on each pass. 2. Execute loop body and apply state modifications. 3. Terminate when the guard condition evaluates to false. 4. The final variable state upon loop termination produces Option (A) (2 2 1 1 2).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q51. What will be the output of the following pseudocode?

```
T The concatenation of two lists is to be performed in
O(1) time. Which of the following implementation of a
list should be used?
```

(A) a) Singly linked list
(B) a) Doubly linked list
(C) a) Circular doubly linked list
(D) a) Array implementation of list

Correct Answer: Option (C) : a) Circular doubly linked list

■ **Step-by-Step Logic:** Execution Trace: 1. Initialize loop variables and verify loop condition on each pass. 2. Execute loop body and apply state modifications. 3. Terminate when the guard condition evaluates to false. 4. The final variable state upon loop termination produces Option (C) (a) Circular doubly linked list).

[Campus Monk Day 04 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q52. What will be the output of the following pseudocode?

```
1.     Find the output of the following pseudo code.
Integer x,y,z
Set x=9, y=2, z=1
if ( x+y > 8)
if ( 7> x+z)
z=7
End if
If ( 9> x-z)
z=9
End if
y=y+z+1
End if

Print x+y+z
```

(A) 31
(B) 11
(C) 42
(D) 30

Correct Answer: Option (D) : 30

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches Option (D) (30).

[Campus Monk Day 06 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q53. How many times campus monk will be printed? Set k = 1, p = 4 Int j = p + k / 2 If (j equals p) Print "campus monk" End if j = p + k / 5 If (j not equal p) Print "campus monk" End if Print "campus monk"

(A) 1 Times
(B) 2 Times
(C) 3 Times
(D) 4 Times

Correct Answer: Option (B) : 2 Times

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches Option (B) (2 Times).

[Campus Monk Day 06 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q54. What will be the output of the following pseudocode? Initial values: a = 4, b = 4, c = 7 Int fun(int a, int b, int c) { For (each c from 3 to 5) { a = (c + c) ^ b; If ((a + c) < (c + a)) { b = (a + 11) + c; } Else { c = b + b; a = 3 + b; continue; } } Return a + b; }

(A) 10
(B) 12
(C) Error
(D) 11

Correct Answer: Option (D) : 11

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (D) (11).

[Campus Monk Day 07 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q55. What will be the output of the following pseudocode?

```
Predict the output of the given pseudocode if the value
of the number is 6: Read number k = 2 i = 2 while i <=
number: k = k * i i = i + 1 end while write k
```

(A) 1440.0
(B) 1340.0
(C) 1700.0
(D) 1560.0

Correct Answer: Option (A) : 1440.0

■ **Step-by-Step Logic:** Execution Trace: 1. Initialize loop variables and verify loop condition on each pass. 2. Execute loop body and apply state modifications. 3. Terminate when the guard condition evaluates to false. 4. The final variable state upon loop termination produces Option (A) (1440.0).

[Campus Monk Day 08 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q56. What will be the output of the following pseudocode?

```
Q1.) Integer p, q, r Set p = 0, q = 2, r = 9 r = 7 + p q
= q + r for (each r from 4 to 7) p = (p + p) & q if ((p +
q) < (r - p) || 8 < p) p = (p + 2) + q Jump out of the
loop End if End for Print p + q
```

(A) 5
(B) 9
(C) 19
(D) 20

Correct Answer: Option (B) : 9

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (B) (9).

[Campus Monk Day 08 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q57. What will be the output of the following pseudocode?

```
Q3.) Integer value, n Set value = 1, n = 45 while (value less than equal to n) value = value << 1 end loop Print value
```

(A) 16
(B) 32
(C) 64
(D) 36

Correct Answer: Option (C) : 64

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to Option (C) (64).

[Campus Monk Day 08 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q58. What will be the output of the following pseudocode?

```
Q6. What is the output for p=15 and q=4. Integer solve (Integer p, Integer q) Integer value while(q) value = p MOD q p = q q = value End while return p End function solve()
```

(A) 3
(B) 1
(C) 2
(D) 0

Correct Answer: Option (B) : 1

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields Option (B) (1).

[Campus Monk Day 08 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q59. What will be the output of the following pseudocode?

```
Q9. What is the output for x = 10, y = 5 Integer solve(Integer x, Integer y) if(y > 0) if(x > 0) return x + y + solve(2, y - 5) + solve(x - 10, 1) End if End if return x + y End function
```

(A) undefined

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields .

[Campus Monk Day 08 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q60. What will be the output of the following pseudocode?

```
Q10. What is the output of following Integer value, n Set value = 32, n = 1 while(value >= n) value = value >> 1 end loop Print value
```

(A) undefined

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to .

[Campus Monk Day 09 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q61. What will be the output of the following pseudocode?

```
What is the output of following pseudo code: Fun() { Int a=10 , b=20 , c=30; If(c>b>a) Print("True"); Else Print("False"); return 0; }
```

(A) True
(B) False
(C) Error
(D) -1

Correct Answer: Option (B) : False

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields Option (B) (False).

[Campus Monk Day 09 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q62. What will be the output of the following pseudocode?

```
What is the output of following: Integer x, y Set x = 15, y = 12 y = x – 1 do{ Print x x = y + (x – 2) } while(x < 40) end do while
```

(A) 14 26
(B) 15 27 39
(C) 27 39
(D) 15

Correct Answer: Option (B) : 15 27 39

■ **Step-by-Step Logic:** Execution Trace: 1. Initialize loop variables and verify loop condition on each pass. 2. Execute loop body and apply state modifications. 3. Terminate when the guard condition evaluates to false. 4. The final variable state upon loop termination produces Option (B) (15 27 39).

[Campus Monk Day 09 Mock Test]

Topic: Campus Monk Mock Pseudocode

Q63. What will be the output of the following pseudocode?

```
What is the output if fun(5) is called? Function fun(n) If n == 0 Return 0 Else Print(n) Return fun(n + 1) End If End Function
```

(A) 1 2 3 4 5
(B) 5 4 3 2 1
(C) Stack overflow
(D) -1

Correct Answer: Option (C) : Stack overflow

■ **Step-by-Step Logic:** Execution Trace: 1. Identify base cases for termination. 2. Trace recursive function call stack and branch expansions. 3. Evaluate leaf returns and unwind the call stack bottom-up. 4. Summing/accumulating the return values yields Option (C) (Stack overflow).

[College ITM Mock Paper 1]

Topic: Infosys SE College Mock Test

Q64. What will be the output of the following pseudocode?

```
What is printed?
x = 45; count = 0;
while (x != 0) {
        x = x & (x - 1);
        count++;
}
print(count ^ (45 >> 2))
```

(A) 4
(B) 11
(C) 15
(D) 7

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to .

[College ITM Mock Paper 1]

Topic: *Infosys SE College Mock Test*

Q65. What will be the output of the following pseudocode?

```
What is printed?
s = 0
for i = 1 to 5
        for j = i to 5
                if ((i + j) % 3 == 0) continue
                if (j == 5) break
                s = s + i * j
        end for
end for
print(s)
```

(A) 38
(B) 42
(C) 46
(D) 50

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Initialize loop variables and verify loop condition on each pass. 2. Execute loop body and apply state modifications. 3. Terminate when the guard condition evaluates to false. 4. The final variable state upon loop termination produces .

[College ITM Mock Paper 2]

Topic: *Infosys SE College Mock Test*

Q66. What will be the output of the following pseudocode?

```
What is printed? (x and y are unsigned integers)
x = 58; y = 0;
while (x > 0) {
        y = (y << 1) | (x & 1);
        x = x >> 1;
}
print(y)
```

(A) 23
(B) 29
(C) 37
(D) 41

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to .

[College ITM Mock Paper 2]

Topic: *Infosys SE College Mock Test*

Q67. What will be the output of the following pseudocode?

```
What is printed?
s = 0
for i = 1 to 4
        for j = 1 to 4
                if (i == j) continue
                if (i + j > 6) break
                s = s + (i * j)
        end for
end for
print(s)
```

(A) 40
(B) 44
(C) 46
(D) 48

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Initialize loop variables and verify loop condition on each pass. 2. Execute loop body and apply state modifications. 3. Terminate when the guard condition evaluates to false. 4. The final variable state upon loop termination produces .

[College ITM Mock Paper 2]

Topic: *Infosys SE College Mock Test*

Q68. What will be the output of the following pseudocode?

```
A queue is used (FIFO). What is printed?
enqueue(2); enqueue(4); enqueue(6);
a = dequeue();
enqueue(a + 10);
b = dequeue();
c = dequeue();
print(b + c)
```

(A) 10
(B) 16
(C) 18
(D) 22

Correct Answer: Option (A) : 10

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches Option (A) (10).

[College ITM Mock Paper 3]

Topic: *Infosys SE College Mock Test*

Q69. What will be the output of the following pseudocode?

```
A stack is used (LIFO). What is printed?
```

(A) 39
(B) 33
(C) 42
(D) 30

Correct Answer: Option (A) : 39

■ **Step-by-Step Logic:** Execution Trace: 1. Trace variable allocations and sequential statement assignments. 2. Evaluate conditional branches and execute the active branch. 3. Compute final expression for display. 4. The evaluated result matches Option (A) (39).

[College ITM Mock Paper 4]

Topic: *Infosys SE College Mock Test*

Q70. What will be the output of the following pseudocode?

```
This algorithm reverses the first k elements of a queue
using a stack. What is printed?
queue = [1, 2, 3, 4, 5]
stack = []
k = 3
for i = 1 to k
        push(stack, dequeue(queue))
end for
while (stack is not empty)
        enqueue(queue, pop(stack))
end while
for i = 1 to (5 - k)
        enqueue(queue, dequeue(queue))
end for
print(queue)
```

(A) [3, 2, 1, 4, 5]
(B) [1, 2, 3, 4, 5]
(C) [5, 4, 3, 2, 1]
(D) [3, 2, 1, 5, 4]

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Initialize loop variables and verify loop condition on each pass. 2. Execute loop body and apply state modifications. 3. Terminate when the guard condition evaluates to false. 4. The final variable state upon loop termination produces .

[College ITM Mock Paper 5]

Topic: Infosys SE College Mock Test

Q71. What will be the output of the following pseudocode?

```
What is printed? (>>=right shift, <<=left shift, &=AND,
^=XOR)
x = 0;
for i = 0 to 7
        if (((45 >> i) & 1) == 1)
                x = x + (1 << (7 - i))
        end if
end for
y = x ^ 45
print(y)
```

(A) 153
(B) 180
(C) 135
(D) 171

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Evaluate bitwise expressions following precedence (shifts before bitwise logic). 2. Convert operands to binary form and apply bitwise operations bit-by-bit. 3. Perform remaining arithmetic updates on the accumulated values. 4. The final printed output evaluates to .

[College ITM Mock Paper 6]

Topic: Infosys SE College Mock Test

Q72. What will be the output of the following pseudocode?

```
What is printed?
s = 0
for i = 1 to 5
        for j = 1 to 5
                if ((i + j) % 4 == 0) continue
                if (j > i + 1) break
                s = s + i * j
        end for
end for
print(s)
```

(A) 142
(B) 138
(C) 146
(D) 150

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Initialize loop variables and verify loop condition on each pass. 2. Execute loop body and apply state modifications. 3. Terminate when the guard condition evaluates to false. 4. The final variable state upon loop termination produces .

[College ITM Mock Paper 6]

Topic: Infosys SE College Mock Test

Q73. The following sorts a stack using a temporary stack (the top of a stack is its last element). How many

```
times is an element moved from tmp back to st?
st = [3, 1, 4, 2]; tmp = []; moves = 0
while (st is not empty)
        t = pop(st)
        while (tmp is not empty and top(tmp) > t)
                push(st, pop(tmp))
                moves = moves + 1
        end while
        push(tmp, t)
end while
print(moves)
```

(A) 3
(B) 2
(C) 4
(D) 5

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Initialize loop variables and verify loop condition on each pass. 2. Execute loop body and apply state modifications. 3. Terminate when the guard condition evaluates to false. 4. The final variable state upon loop termination produces .

[College ITM Mock Paper 6]

Topic: Infosys SE College Mock Test

Q74. What will be the output of the following pseudocode?

```
What is printed? (dynamic programming for the minimum
number of coins; INF is a very large number)
coins = [1, 3, 4]
dp[0] = 0; dp[1..6] = INF
for i = 1 to 6
        for each c in coins
                if (i >= c)
                        dp[i] = min(dp[i], dp[i - c] + 1)
                end if
        end for
end for
print(dp[6])
```

(A) 2
(B) 3
(C) 4
(D) 6

Correct Answer: See Logic

■ **Step-by-Step Logic:** Execution Trace: 1. Initialize loop variables and verify loop condition on each pass. 2. Execute loop body and apply state modifications. 3. Terminate when the guard condition evaluates to false. 4. The final variable state upon loop termination produces .